

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

ME 623 — Cryogenic Engineering II · Autumn 2026

A General Solver for Precooled Joule–Thomson Cryocoolers

Reproduction, validation and extension of a 1 K He-3 hybrid cryocooler model

Presented by

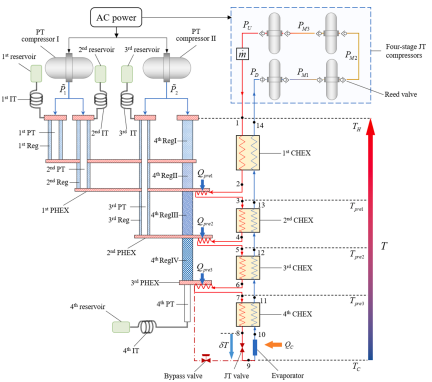
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Reference paper. T. Zhang and H. Dang, "Investigations on a 1 K hybrid cryocooler composed of a four-stage Stirling-type pulse tube cryocooler and a Joule–Thomson cooler. Part A: Theoretical analyses and modeling", *Cryogenics* **116** (2021) 103282. doi:10.1016/j.cryogenics.2021.103282

Motivation and reference work



Hybrid cryocooler; Zhang & Dang (2021), Fig. 1.

Need. Detectors such as SNSPDs require $\sim 1\,\mathrm{K}$. Stirling-type pulse-tube cryocoolers (SPTC) are limited to $\sim 3\,\mathrm{K}$; Joule–Thomson (JT) expansion reaches 1 K provided the working fluid is precooled.

System. Four-stage SPTC coupled to a He-3 JT loop: four counterflow heat exchangers (CHEX), three precooling heat exchangers (PHEX) on the SPTC stages, an orifice of $d = 20\,\mu\mathrm{m}$ and an evaporator at $T_C = 1\,\mathrm{K}$, $P_D = P_{\mathrm{sat}}(T_C) = 1.16\,\mathrm{kPa}$.

Reported optimum (last stage)

$$P_{U,\mathrm{opt}} = 0.36\,\mathrm{MPa},\ T_{\mathrm{pre3}} = 8\,\mathrm{K},\ q_C = 8.0\,\mathrm{kJ/kg}$$
$$\dot{m} = 2.34\,\mathrm{mg/s},\ Q_C = 18.7\,\mathrm{mW}$$

Objectives

1. **Analysis.** Re-derive the enthalpy-flow model of the reference paper and state all of its assumptions.

2. **Solver.** Formulate a general solver for the complete cycle: all 14 states, an arbitrary number of stages, any working fluid, and the load on every precooler.

3. **Validation.** Reproduce the published figures and values; then apply the same solver, with their own inputs only, to published coolers of the same architecture.

4. **Design.** Determine what the reference paper leaves open: the temperatures and loads required of each precooling stage.

Enthalpy-flow model and assumptions

Counterflow exchanger (hot $A \to B$ at P_U, cold $C \to D$ at P_D):

$$h_A - h_B = h_D - h_C \equiv Q$$

Precooler j fixes $T_3, T_5, T_7 = T_{\mathrm{pre},j}$; its load is a result:

$$q_{\mathrm{pre},1} = h_2 - h_3, \quad q_{\mathrm{pre},2} = h_4 - h_5, \quad q_{\mathrm{pre},3} = h_6 - h_7$$

Valve and evaporator:

$$h_9 = h_8, \qquad q_C = h_{10} - h_8, \qquad Q_C = \dot{m}\, q_C$$

Global balance (paper Eq. 22, $\eta = 1$):

$$q_C = \sum_j q_{\mathrm{pre},j} + \left[h(P_D, T_H) - h(P_U, T_H)\right]$$

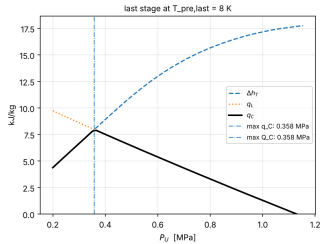
Assumptions

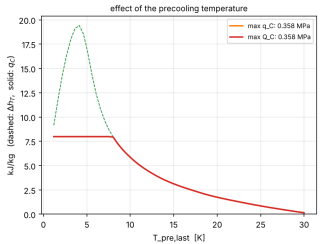
- Steady state; equal mass flow in both streams
- No pressure drop: states 1–8 at P_U, 9–14 at $P_D = P_{\mathrm{sat}}(T_C)$
- No heat in-leak or axial conduction
- Precooler outlet temperatures prescribed
- Each CHEX characterised by an effectiveness η
- Real-fluid He-3 properties

The last exchanger: two limits

$$q_C = (h_{10} - h_7) + Q, \qquad Q \le \min\Big(\underbrace{h(P_D, T_7) - h_{10}}_{\text{warm-end limit}},\ \underbrace{h_7 - h(P_U, T_C)}_{\text{cold-end limit}}\Big)$$

$$\Longrightarrow \quad \boxed{q_C = \min\left(\Delta h_T,\, q_L\right)}, \quad \Delta h_T = h(P_D, T_{\text{pre}}) - h(P_U, T_{\text{pre}}), \quad q_L = h_g(T_C) - h(P_U, T_C)$$





- Δh_T increases with P_U below the inversion temperature; q_L decreases and vanishes at 1.13 MPa.
- Optimum where both ends pinch: $\Delta h_T = q_L$ (paper Eq. 34).
- 8 K is the warmest T_{pre} on the q_L plateau.
- For $\eta < 1$: $q_C = \eta \min(\Delta h_T, q_L) - (1-\eta)(h_7 - h_{10})$, with $h_7 - h_{10} \approx 41\,\text{kJ/kg} \approx 5\,q_C$.

Structure of the solver

1 Properties

He-3 tables (0.5 K to 300 K, 1 Pa–3 MPa) implemented as a property library: $h(P,T)$, $T(P,h)$, s, ρ and saturation data. Monotone (PCHIP) interpolation; liquid and vapour branches kept separate; no extrapolation. Other fluids via CoolProp.

2 Cycle

The evaporator outlet (saturated vapour at T_C) is known. Exchangers are solved from CHEX n to CHEX 1, each with both inlets known:

$$Q = \eta \, \min(W_{\text{warm}}, W_{\text{cold}}),$$
$$h_B = h_A - Q, \qquad h_D = h_C + Q.$$

Explicit; no iteration. Precooler loads follow from enthalpy differences.

3 Orifice flow (HEM)

Along the isentrope from state 8:

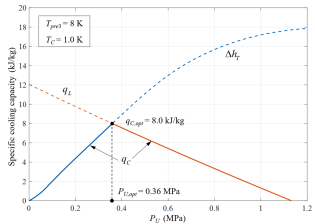
$$G(P) = \rho\sqrt{2\,(h_0 - h)},$$
$$\dot{m} = \tfrac{\pi d^2}{4} \max_P G.$$

$\mathrm{d}G/\mathrm{d}P = 0 \iff u = a$: the maximum is the choked flux, obtained without the speed of sound.

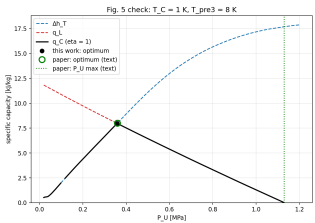
Outputs. All states; q_C, $\dot{m}$, Q_C; every $Q_{\text{pre},j}$; optimal P_{LI} and T_{pre}.

Checks. Global energy balance, internal-pinch scan, feasibility of the inputs.

Specific cooling capacity versus upstream pressure



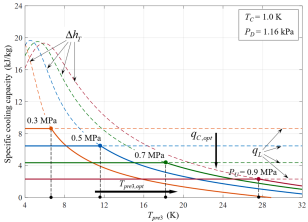
(a) Zhang & Dang (2021), Fig. 5



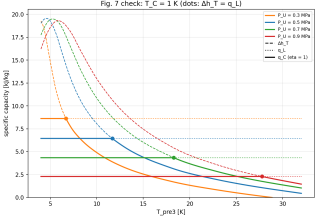
(b) This work, identical inputs

$T_C = 1\,\mathrm{K}, T_{\mathrm{pre3}} = 8\,\mathrm{K}$	This work	Paper	Difference
$P_{U,\mathrm{opt}}$ [MPa]	0.358	0.36	−0.4%
$q_{C,\mathrm{opt}}$ [kJ/kg]	7.98	8.0	−0.3%
$P_{U,\mathrm{max}}$ ($q_L = 0$) [MPa]	1.131	1.13	+0.1%
$\max \lvert q_C - \min(\Delta h_T, q_L) \rvert / q_C$	4×10^{-14}	—	identity
$\lvert \Delta h_T - q_L \rvert / q_L$ at optimum	7×10^{-8}	—	Eq. 34

Effect of the precooling temperature



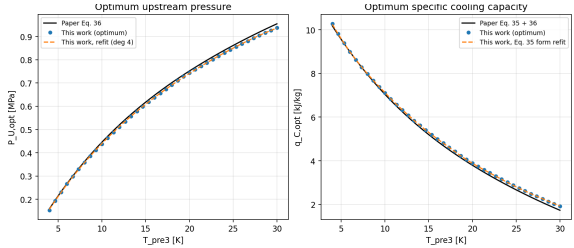
(a) Zhang & Dang (2021), Fig. 7



(b) This work, identical inputs

P_{LI} [MPa]	$T_{\mathrm{pre3,opt}}$, this work [K]	Paper, Eq. 36 [K]	Difference [K]
0.3	6.69	6.67	0.01
0.5	11.67	11.44	0.23
0.7	18.28	17.94	0.34
0.9	27.76	26.99	0.78

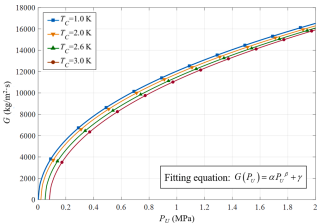
Optimum correlations (paper Eqs. 35–36)



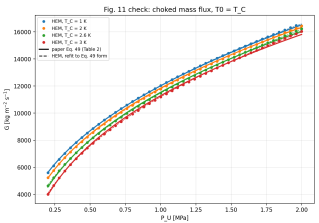
This work, $T_C = 1\,\mathrm{K}$, $\eta = 1$: optimum at 40 values of T_{pre3} (points). Right: $q_{C,\mathrm{opt}}$ at $P_U = P_{U,\mathrm{opt}}(T_{\mathrm{pre3}})$; paper curve is Eq. 35 combined with Eq. 36.

$T_{\mathrm{pre3}} = 4$–$30\,\mathrm{K}$	Mean difference	Maximum difference
$P_{U,\mathrm{opt}}$ vs. Eq. 36	1.39%	3.72%
$q_{C,\mathrm{opt}}$ vs. Eqs. 35 and 36 combined	3.12%	10.4% (at 30 K)
$q_{C,\mathrm{opt}}$ vs. Eq. 35 at the computed $P_{U,\mathrm{opt}}$	0.17%	0.47%

Orifice mass flux (paper Eq. 49, Table 2)



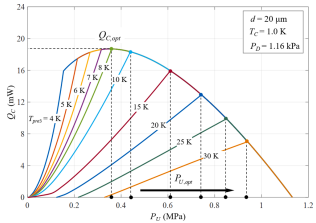
(a) Zhang & Dang (2021), Fig. 11



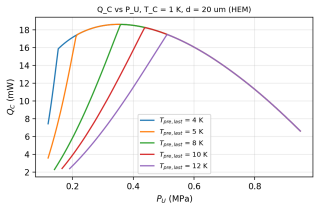
(b) This work, identical inputs

$T_0 = T_C$ [K]	HEM vs. Eq. 49, mean	maximum	Eq. 49 form fitted to HEM, maximum
1.0	0.25%	0.71%	0.24%
2.0	0.41%	0.72%	0.53%
2.6	0.51%	2.28%	1.00%
3.0	0.69%	1.49%	1.82%

Gross cooling capacity versus upstream pressure



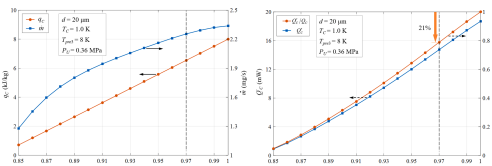
(a) Zhang & Dang (2021), Fig. 13



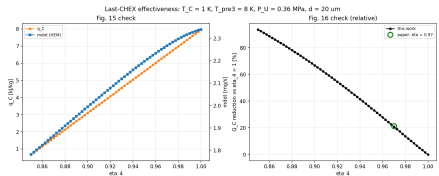
(b) This work, identical inputs

$d = 20\,\mu\text{m}$, $P_U = 0.36\,\text{MPa}$, $T_{\text{pre3}} = 8\,\text{K}$, $\eta = 1$	This work	Paper	Difference
q_C [kJ/kg]	7.96	8.0	−0.5%
$\dot{m}$ [mg/s]	2.339	2.34	−0.04%
Q_C [mW]	18.62	18.7	−0.4%

Effectiveness of the last exchanger



(a) Zhang & Dang (2021), Figs. 15–16



(b) This work, identical inputs

η_4	T_8 [K]	q_C [kJ/kg]	$\dot{m}$ [mg/s]	Q_C [mW] (paper)
1.00	1.00	7.96	2.339	18.62 (18.7)
0.97	1.91	6.50	2.267	14.74 (14.7)
0.91	2.96	3.59	2.036	7.31
0.85	3.62	0.67	1.781	1.20

Reduction of Q_C at $\eta_4 = 0.97$: **20.8%** (paper $\approx 21\%$).

Sensitivity: $\approx 6.9\%$ of Q_C per 0.01 of η_4 (paper $\approx 7\%$).

Summary of the validation checks

Source	Quantity	This work	Paper	Tolerance
Fig. 5	$P_{U,\mathrm{opt}}$ [MPa]; $q_{C,\mathrm{opt}}$ [kJ/kg]; $P_{U,\max}$ [MPa]	0.358; 7.98; 1.131	0.36; 8.0; 1.13	± 0.03; ± 0.4; ± 0.05
	$q_C = \min(\Delta h_T, q_L)$ (rel. deviation); optimum inside search range	4×10^{-14}; yes	—	10^{-6}
Eqs. 35–36	$\Delta h_T = q_L$ at optimum (rel. deviation)	7.5×10^{-8}	—	10^{-2}
	$P_{U,\mathrm{opt}}$ vs. Eq. 36: mean; maximum	1.39%; 3.72%	—	5%; 10%
	$q_{C,\mathrm{opt}}$ vs. Eqs. 35–36: mean; optima on search bound	3.12%; 0	—	5%; 0
Fig. 7	$T_{\mathrm{pre3,opt}}$ vs. Eq. 36 at 0.3/0.5/0.7/0.9 MPa [K]	0.01/0.23/0.34/0.78	—	± 1.5
Eq. 49	HEM vs. Table 2, mean, $T_C = 1/2/2.6/3$ K	0.25/0.41/0.51/0.69%	—	15%
	Eq. 49 form fitted to HEM, maximum	0.24/0.53/1.00/1.82%	—	2%
Figs. 15–16	q_C; Q_C; $\dot{m}$ at $\eta = 1$	7.96; 18.62; 2.339	8.0; 18.7; 2.34	± 0.4; ± 1; ± 0.15
	Q_C at $\eta = 0.97$ [mW]; reduction [%]	14.74; 20.8	14.7; 21.4	± 1; ± 3

Application to other coolers (He-4)

Both works are cited by Zhang & Dang and report ideal-model values (calculated or design values), which are the appropriate comparison for an ideal solver. Only the inputs given in each paper are used.

	Liu et al. (2017) [1]	Ma et al. (2019) [2], design point
T_C [K]	4.25	4.5
P_U [MPa]	1.137	2.0
T_{pre}, last / upper [K]	17.3	15 / 100
$\dot{m}$ [mg/s]	9.1	10.5
η, last CHEX	0.9857*	$\geq$ 0.97 (stated)
Q_C, reported [mW]	52.5 (calculated)	100 (design)
Q_C, this work [mW]	**53.1** (+1.1%)	**90.2–108.8** ($\eta = 0.97$–1)
q_C, this work / reported [kJ/kg]	5.84 / 5.77	100 mW $\Rightarrow \eta \approx 0.985$
Q_{pre} at 15 K, this work / design [mW]	—	278 / 260

*Derived from the reported 17.3 K inlet and 17.1 K return-outlet temperatures; not fitted.

[1] D. Liu, Z. Gan, A.T.A.M. de Waele, X. Tao, Y. Yao, *Int. J. Heat Mass Transf.* **109** (2017) 1094–1099. doi:10.1016/j.ijheatmasstransfer.2017.02.070

[2] Y. Ma, J. Quan, J. Wang, Y. Liu, J. Li, J. Liang, *Cryogenics* **104** (2019) 102983. doi:10.1016/j.cryogenics.2019.102983

Sizing the precooling stages

1. Fix the cold end (T_C, d) and determine the last-stage optimum $P_{U,\mathrm{opt}}$ and $T^{\mathrm{opt}}_{\mathrm{pre},n-1}$.

2. The last CHEX depends only on $(P_U, T_{\mathrm{pre},n-1})$ and $h_g(T_C)$; the upper precooling temperatures therefore **do not affect** q_C, $\dot{m}$ or Q_C, and only distribute a fixed total load (Eq. 22):
$$\sum_j q_{\mathrm{pre},j} = q_C - \big[h(P_D, T_H) - h(P_U, T_H)\big].$$

3. Sweep the upper precooling temperatures $T_1 > T_2 > \cdots > T_{n-2}$ and record all loads:
$$(T_1, \ldots, T_{n-2}) \longmapsto (q_{\mathrm{pre},1}, \ldots, q_{\mathrm{pre},n-1}), \qquad Q_{\mathrm{pre},j} = \dot{m}\, q_{\mathrm{pre},j}.$$

4. The resulting load map is the interface to the cryocooler: **any optimiser** for the Stirling design (stage capacities, efficiency, cost) can operate on it.

Example: minimum Carnot-weighted precooling load

Minimise the ideal work required to reject each precooler load from T_j to $T_H = 300\,\mathrm{K}$:

$$\min_{T_1 > \cdots > T_{n-2}} W = \sum_{j=1}^{n-1} Q_{\mathrm{pre},j} \left(\frac{T_H}{T_j} - 1 \right) \qquad \text{subject to } Q_{\mathrm{pre},j} \geq 0.$$

He-3, $T_C = 1\,\mathrm{K}$, $d = 20\,\mu\mathrm{m}$, $\eta = 1$	PHEX 1	PHEX 2	PHEX 3	Last stage
Temperature [K]	25.6	12.3	8.0	$P_{LI} = 0.358\,\mathrm{MPa}$ (computed)
Load, minimum W [mW]	9.9	8.3	8.6	$Q_C = 18.62\,\mathrm{mW}$
Load, temperatures 80/25/8 K [mW]	0.5	9.7	16.6	$Q_C = 18.62\,\mathrm{mW}$

The total precooling load is 26.8 mW in both cases; with the optimised distribution the 8 K stage carries half the load.

Conclusions

- **Theory.** With $q_C = (h_{10} - h_7) + Q$ and the two limits of a counterflow exchanger, the published result $q_C = \min(\Delta h_T, q_L)$ and the optimum condition $\Delta h_T = q_L$ follow directly: at the optimum both ends of the last exchanger pinch.
- **Validation.** A single general solver reproduces Zhang & Dang to within about 1% (optimum, correlations, mass flux, Q_C, effectiveness), the calculated Q_C of Liu et al. to within 1.1%, and brackets the design value of Ma et al.
- **Design.** The upper precoolers only distribute a fixed load; sweeping their temperatures yields a load map on which any cryocooler optimisation can be performed.
- **Limitations.** Ideal model without heat in-leak or pressure drop; exchanger effectiveness is an input; He-3 property data cover 0.5–3.2 K.